

Basics of FE-Analysis

BK10A6400

Lecture 5: 2D solid (plane) elements (Cook's book chapter 3, Hakala p. 260)

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ANSYS ↔ FEMAP / NX Nastran element equivalence

Type / Application	ANSYS	FEMAP / NX Nastran
Rod (axial only)	LINK180	CROD
Beam (Timoshenko)	BEAM188 / BEAM189	CBEAM / CBAR
2D solid (linear)	PLANE42 (old)	CQUAD4 / CTRIA3
2D solid (linear)	PLANE182	CQUAD4 / CTRIA3
2D solid (quadratic)	PLANE183	CQUAD8 / CTRIA6
2D sxisymmetric	PLANE182 / 183 (KEYOPT=3)	CAX4 / CAX8
Shell / Plate	SHELL181	CQUAD4 / CQUAD8 (Shell)
3D solid	SOLID185	CHEXA
3D solid	SOLID186	CPENTA
3D solid	SOLID187	CTETRA

CROD for axial only, CBEAM/CBAR for full 3D beam, plane stress/strain via Property (FEMAP) or KEYOPT (ANSYS).

Example: A simple cantilever problem solved with one 4 DOF beam element: completed together in the lecture

RECAP: The elemental stiffness matrix for a 2D Euler-Bernoulli 4 DOF beam element:

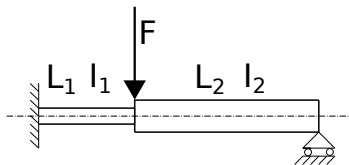
$$\overline{\mathbf{K}}_{EB4DOF} = \begin{pmatrix} \frac{12EI}{L^3} & \frac{6EI}{L^2} & -\frac{12EI}{L^3} & \frac{6EI}{L^2} \\ \frac{6EI}{L^2} & \frac{4EI}{L} & -\frac{6EI}{L^2} & \frac{2EI}{L} \\ -\frac{12EI}{L^3} & -\frac{6EI}{L^2} & \frac{12EI}{L^3} & -\frac{6EI}{L^2} \\ \frac{6EI}{L^2} & \frac{2EI}{L} & -\frac{6EI}{L^2} & \frac{4EI}{L} \end{pmatrix}$$

Example: A simple cantilever problem ...

Example: A simple cantilever problem ...

Example: Beam structure with two beam elements

The aim is to solve nodal displacements and member forces using a two-dimensional **two-noded Euler-Bernoulli beam element (6 DOF)** in MATLAB. For simplicity, a mesh of two beam elements is only used to analyze the beam structure shown in the figure.



- Discretized by two 6 DOF's E-B elements
- 3 nodes, 3 DOF's at node, $3 \times 3 = 9$ DOF's
- 4 fixed/constrained nodal displacements.

Elemental stiffness matrices

Element 1:

$$\overline{\mathbf{K}}^{(1)} = \begin{matrix} & \begin{matrix} u_1 & v_1 & \theta_1 & u_2 & v_2 & \theta_2 \end{matrix} \\ \begin{pmatrix} \frac{E_1 A_1}{L_1} & 0 & 0 & -\frac{E_1 A_1}{L_1} & 0 & 0 \\ 0 & \frac{12 E_1 I_1}{L_1^3} & \frac{6 E_1 I_1}{L_1^2} & 0 & -\frac{12 E_1 I_1}{L_1^3} & \frac{6 E_1 I_1}{L_1^2} \\ 0 & \frac{6 E_1 I_1}{L_1^2} & \frac{4 E_1 I_1}{L_1} & 0 & -\frac{6 E_1 I_1}{L_1^2} & \frac{2 E_1 I_1}{L_1} \\ -\frac{E_1 A_1}{L_1} & 0 & 0 & \frac{E_1 A_1}{L_1} & 0 & 0 \\ 0 & -\frac{12 E_1 I_1}{L_1^3} & -\frac{6 E_1 I_1}{L_1^2} & 0 & \frac{12 E_1 I_1}{L_1^3} & -\frac{6 E_1 I_1}{L_1^2} \\ 0 & \frac{6 E_1 I_1}{L_1^2} & \frac{2 E_1 I_1}{L_1} & 0 & -\frac{6 E_1 I_1}{L_1^2} & \frac{4 E_1 I_1}{L_1} \end{pmatrix} & \begin{matrix} u_1 \\ v_1 \\ \theta_1 \\ u_2 \\ v_2 \\ \theta_2 \end{matrix} \end{matrix}$$

Elemental stiffness matrices

Element 2:

$$\overline{\mathbf{K}}^{(2)} = \begin{matrix} & \begin{matrix} u_2 & v_2 & \theta_2 & u_3 & v_3 & \theta_3 \end{matrix} \\ \begin{pmatrix} \frac{E_2 A_2}{L_2} & 0 & 0 & -\frac{E_2 A_2}{L_2} & 0 & 0 \\ 0 & \frac{12E_2 I_2}{L_2^3} & \frac{6E_2 I_2}{L_2^2} & 0 & -\frac{12E_2 I_2}{L_2^3} & \frac{6E_2 I_2}{L_2^2} \\ 0 & \frac{6E_2 I_2}{L_2^2} & \frac{4E_2 I_2}{L_2} & 0 & -\frac{6E_2 I_2}{L_2^2} & \frac{2E_2 I_2}{L_2} \\ -\frac{E_2 A_2}{L_2} & 0 & 0 & \frac{E_2 A_2}{L_2} & 0 & 0 \\ 0 & -\frac{12E_2 I_2}{L_2^3} & -\frac{6E_2 I_2}{L_2^2} & 0 & \frac{12E_2 I_2}{L_2^3} & -\frac{6E_2 I_2}{L_2^2} \\ 0 & \frac{6E_2 I_2}{L_2^2} & \frac{2E_2 I_2}{L_2} & 0 & -\frac{6E_2 I_2}{L_2^2} & \frac{4E_2 I_2}{L_2} \end{pmatrix} & \begin{matrix} u_2 \\ v_2 \\ \theta_2 \\ u_3 \\ v_3 \\ \theta_3 \end{matrix} \end{matrix}$$

The transformation from a local coordinate system to a global coordinate system. Now $\theta = 0$ so the transformation is not necessarily needed because $\mathbf{T}$ is a unit matrix then.

$$\mathbf{K}_G^{(1)} = \mathbf{T}^T(0) \overline{\mathbf{K}}^{(1)} \mathbf{T}(0) \quad ; \quad \mathbf{K}_G^{(2)} = \mathbf{T}^T(0) \overline{\mathbf{K}}^{(2)} \mathbf{T}(0) \quad (1)$$

Assembling

$$K_G = \begin{matrix} & \begin{matrix} u_1 & v_1 & \theta_1 & u_2 & v_2 & \theta_2 & u_3 & v_3 & \theta_3 \end{matrix} \\ \begin{matrix} k_{11}^{(1)} \\ k_{12}^{(1)} \\ k_{21}^{(1)} \\ k_{22}^{(1)} \\ k_{31}^{(1)} \\ k_{32}^{(1)} \\ k_{41}^{(1)} \\ k_{51}^{(1)} \\ k_{61}^{(1)} \\ 0 \\ 0 \\ 0 \end{matrix} & \begin{matrix} k_{12}^{(1)} \\ k_{22}^{(1)} \\ k_{32}^{(1)} \\ k_{42}^{(1)} \\ k_{52}^{(1)} \\ k_{62}^{(1)} \\ 0 \\ 0 \\ 0 \end{matrix} & \begin{matrix} k_{13}^{(1)} \\ k_{23}^{(1)} \\ k_{33}^{(1)} \\ k_{43}^{(1)} \\ k_{53}^{(1)} \\ k_{63}^{(1)} \\ 0 \\ 0 \\ 0 \end{matrix} & \begin{matrix} k_{14}^{(1)} \\ k_{24}^{(1)} \\ k_{34}^{(1)} \\ k_{44}^{(1)} + k_{11}^{(2)} \\ k_{54}^{(1)} + k_{21}^{(2)} \\ k_{64}^{(1)} + k_{31}^{(2)} \\ k_{41}^{(2)} \\ k_{51}^{(2)} \\ k_{61}^{(2)} \end{matrix} & \begin{matrix} k_{15}^{(1)} \\ k_{25}^{(1)} \\ k_{35}^{(1)} \\ k_{45}^{(1)} + k_{12}^{(2)} \\ k_{55}^{(1)} + k_{22}^{(2)} \\ k_{65}^{(1)} + k_{32}^{(2)} \\ k_{42}^{(2)} \\ k_{52}^{(2)} \\ k_{62}^{(2)} \end{matrix} & \begin{matrix} k_{16}^{(1)} \\ k_{26}^{(1)} \\ k_{36}^{(1)} \\ k_{46}^{(1)} + k_{13}^{(2)} \\ k_{56}^{(1)} + k_{23}^{(2)} \\ k_{66}^{(1)} + k_{33}^{(2)} \\ k_{43}^{(2)} \\ k_{53}^{(2)} \\ k_{63}^{(2)} \end{matrix} & \begin{matrix} 0 \\ 0 \\ 0 \\ k_{14}^{(2)} \\ k_{24}^{(2)} \\ k_{34}^{(2)} \\ k_{44}^{(2)} \\ k_{54}^{(2)} \\ k_{64}^{(2)} \end{matrix} & \begin{matrix} 0 \\ 0 \\ 0 \\ k_{15}^{(2)} \\ k_{25}^{(2)} \\ k_{35}^{(2)} \\ k_{45}^{(2)} \\ k_{55}^{(2)} \\ k_{65}^{(2)} \end{matrix} & \begin{matrix} 0 \\ 0 \\ 0 \\ k_{16}^{(2)} \\ k_{26}^{(2)} \\ k_{36}^{(2)} \\ k_{46}^{(2)} \\ k_{56}^{(2)} \\ k_{66}^{(2)} \end{matrix} \end{matrix} \begin{matrix} u_1 \\ v_1 \\ \theta_1 \\ u_2 \\ v_2 \\ \theta_2 \\ u_3 \\ v_3 \\ \theta_3 \end{matrix}$$

Force vector

$$\mathbf{f}_G = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -F \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \begin{matrix} u_1 \\ v_1 \\ \theta_1 \\ u_2 \\ v_2 \\ \theta_2 \\ u_3 \\ v_3 \\ \theta_3 \end{matrix}$$

Elimination of linear constraints

Fixed nodal displacements: $u_1 = v_1 = \theta_1 = v_3 = 0$. Let's eliminate corresponding rows and columns regarding FE-matrices.

$$\mathbf{f}_{G_{red}} = \begin{pmatrix} 0 \\ -F \\ 0 \\ 0 \\ 0 \end{pmatrix} \begin{matrix} u_2 \\ v_2 \\ \theta_2 \\ u_3 \\ \theta_3 \end{matrix}$$

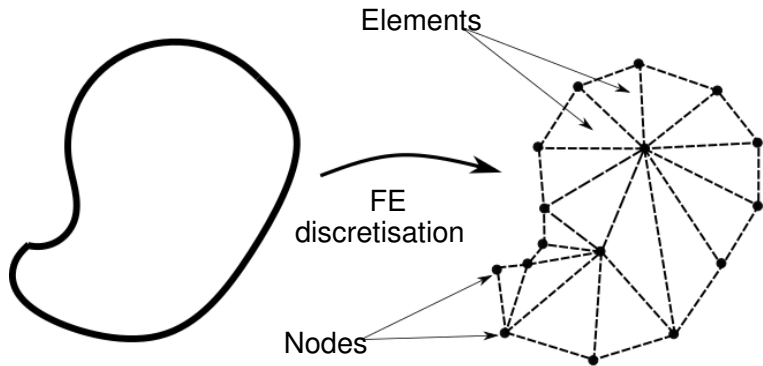
$$\mathbf{K}_{G_{red}} = \begin{pmatrix}
 \begin{matrix} u_2 \\ v_2 \\ \theta_2 \end{matrix} &
 \begin{matrix} u_2 \\ v_2 \\ \theta_2 \end{matrix} &
 \begin{matrix} \theta_2 \\ u_3 \\ \theta_3 \end{matrix} &
 \begin{matrix} u_3 \\ \theta_3 \end{matrix} &
 \begin{matrix} u_3 \\ \theta_3 \end{matrix} \\
 \begin{matrix} k_{44}^{(1)} + k_{11}^{(2)} \\ k_{54}^{(1)} + k_{21}^{(2)} \\ k_{64}^{(1)} + k_{31}^{(2)} \end{matrix} &
 \begin{matrix} k_{45}^{(1)} + k_{12}^{(2)} \\ k_{55}^{(1)} + k_{22}^{(2)} \\ k_{65}^{(1)} + k_{32}^{(2)} \end{matrix} &
 \begin{matrix} k_{46}^{(1)} + k_{13}^{(2)} \\ k_{56}^{(1)} + k_{23}^{(2)} \\ k_{66}^{(1)} + k_{33}^{(2)} \end{matrix} &
 \begin{matrix} k_{14}^{(2)} \\ k_{24}^{(2)} \\ k_{34}^{(2)} \end{matrix} &
 \begin{matrix} k_{16}^{(2)} \\ k_{26}^{(2)} \\ k_{36}^{(2)} \end{matrix} \\
 \begin{matrix} k_{41}^{(2)} \\ k_{51}^{(2)} \\ k_{61}^{(2)} \end{matrix} &
 \begin{matrix} k_{42}^{(2)} \\ k_{52}^{(2)} \\ k_{62}^{(2)} \end{matrix} &
 \begin{matrix} k_{43}^{(2)} \\ k_{53}^{(2)} \\ k_{63}^{(2)} \end{matrix} &
 \begin{matrix} k_{44}^{(2)} \\ k_{54}^{(2)} \\ k_{64}^{(2)} \end{matrix} &
 \begin{matrix} k_{46}^{(2)} \\ k_{56}^{(2)} \\ k_{66}^{(2)} \end{matrix} \\
 u_2 & v_2 & \theta_2 & u_3 & \theta_3
 \end{pmatrix}$$

Note that $\mathbf{K}_{G_{red}}$ is symmetric i.e. $k_{ij} = k_{ji}$ (if not, there is a typo) and $\det \mathbf{K}_{G_{red}} > 0$ (if not, check BC's).

And then we solve displacements (in a global coordinate system).

$$\mathbf{u}_{G_{red}} = \mathbf{K}_{G_{red}}^{-1} \mathbf{f}_{G_{red}}$$

Schematic FE discretisation of an arbitrary 2D structure

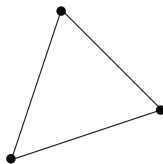


Some two-dimensional solid (plane) elements

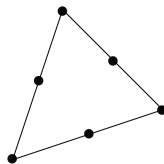
- Constant strain triangle (CST), 3 nodes, 6 DOFs
- Linear strain triangle (LST), 6 nodes, 12 DOFs
- Bilinear quadrilateral (Q4), 4 nodes, 8 DOFs
- 8-node biquadratic quadrilateral (Q8), 8 nodes, 16 DOFs
- 9-node biquadratic quadrilateral (Q9), 9 nodes, 18 DOFs
- Bilinear quadrilateral (improved by "bubble modes"), 4 nodes, 8 DOFs (+ 2 internal DOFs that are eliminated by static condensation).

Some two-dimensional solid (plane) elements

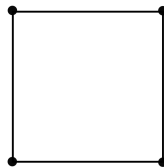
CST



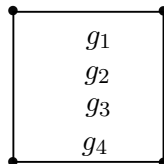
LST



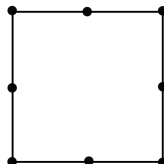
Q4



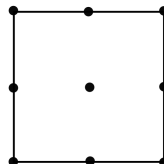
Q6



Q8



Q9



Strains and stresses in 2D

Notations for strains and stresses in a vector form for a 2D case:

$$\boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 2\varepsilon_{12} \end{pmatrix} = \begin{pmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{pmatrix}$$

$$\boldsymbol{\sigma} = \begin{pmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{pmatrix} = \begin{pmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \tau_{xy} \end{pmatrix}$$

Linear relationship (small displacements and rotations)
between strains and displacements:

$$\varepsilon_{xx} = \frac{\partial u_h}{\partial x} \quad ; \quad \varepsilon_{yy} = \frac{\partial v_h}{\partial y} \quad ; \quad \gamma_{xy} = \frac{\partial u_h}{\partial y} + \frac{\partial v_h}{\partial x}$$

Strains in 2D: to be completed in the lecture

- Small angle assumption: $\tan \alpha_1 = \frac{\partial u_h}{\partial y} \approx \alpha_1$,
 $\tan \alpha_2 = \frac{\partial v_h}{\partial x} \approx \alpha_2$

Strains and stresses in 2D

A linear strain-displacement relationship can be written in a matrix form as

$$\underbrace{\begin{pmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{pmatrix}}_{\boldsymbol{\varepsilon}} = \underbrace{\begin{pmatrix} \frac{\partial}{\partial x} & 0 \\ 0 & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{pmatrix}}_{\boldsymbol{D}} \underbrace{\begin{pmatrix} u_h \\ v_h \end{pmatrix}}_{\boldsymbol{u}_h} \quad (2)$$

We remember that

$$\boldsymbol{u}_h = \boldsymbol{N} \boldsymbol{u} \quad ; \quad \boldsymbol{\varepsilon} = \underbrace{\frac{\partial \boldsymbol{N}}{\partial \boldsymbol{x}}}_{\boldsymbol{B}} \boldsymbol{u} \quad (3)$$

where $\boldsymbol{B}$ is so-called the strain-displacement matrix that can be written as

$$\boldsymbol{B} = \begin{pmatrix} \frac{\partial N_1}{\partial x} & 0 & \frac{\partial N_2}{\partial x} & 0 & \frac{\partial N_3}{\partial x} & 0 & \dots \\ 0 & \frac{\partial N_1}{\partial y} & 0 & \frac{\partial N_2}{\partial y} & 0 & \frac{\partial N_3}{\partial y} & \dots \\ \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial x} & \frac{\partial N_2}{\partial y} & \frac{\partial N_2}{\partial x} & \frac{\partial N_3}{\partial y} & \frac{\partial N_3}{\partial x} & \dots \end{pmatrix} \quad (4)$$

Displacement field

Displacement field $\mathbf{u}_h = \{u_h, v_h\}^T$ is interpolated as follows

$$\begin{cases} u_h(x, y) := N_1 u_1 + N_2 u_2 + \dots + N_n u_n \\ v_h(x, y) := N_1 v_1 + N_2 v_2 + \dots + N_n v_n \end{cases} \quad (5)$$

and same in a matrix form:

$$\underbrace{\begin{pmatrix} u_h \\ v_h \end{pmatrix}}_{\mathbf{u}_h} = \underbrace{\begin{pmatrix} N_1 & 0 & N_2 & 0 & \dots & N_n u_n \\ 0 & N_1 & 0 & N_2 & \dots & N_n v_n \end{pmatrix}}_{\mathbf{N}} \underbrace{\begin{pmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ \vdots \\ u_n \\ v_n \end{pmatrix}}_{\mathbf{u}} \quad (6)$$

$$\mathbf{u}_h = \mathbf{N} \mathbf{u} \quad (7)$$

Linear relationship between stresses and strains.

$$\sigma = D\varepsilon \quad (8)$$

It is assumed that a plane is thin (in z -direction) and constant thickness. Then there is no stress through the plate's thickness. In the case of plane stress i.e. $\sigma_{zz} = 0, \sigma_{xz} = 0, \sigma_{yz} = 0$, the elasticity matrix $\mathbf{D}$ for a linear elastic and isotropic material is written as:

$$\mathbf{D} = \frac{E}{1 - \nu^2} \begin{pmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{pmatrix} \quad (9)$$

Plane strain assumptions provided good representations because the in-plane strains are developed from full linear (Hooke's law) 3D elasticity.

- The out-of-plane (z-direction) strains are set to zero (analogy to an object that is constrained by rigid walls in z-direction)
- $\varepsilon_{zz} = \varepsilon_{xz} = \varepsilon_{yz} = 0$

$$\mathbf{D} = \frac{E}{(1+\nu)(1-\nu)} \begin{pmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & \frac{1-2\nu}{2} \end{pmatrix} \quad (10)$$

Strain energy and elemental stiffness matrix

$$W_{int} = \frac{1}{2} \int_V \boldsymbol{\sigma} \boldsymbol{\varepsilon} \, dV = \frac{1}{2} \mathbf{u}^T \underbrace{\int_V \mathbf{B}^T \mathbf{D} \mathbf{B} \, dV}_{\overline{\mathbf{K}}} \mathbf{u} \quad (11)$$

The stiffness matrix can be found via the strain-displacement matrix:

$$\overline{\mathbf{K}} = \int_V \mathbf{B}^T \mathbf{D} \mathbf{B} \, dV \quad (12)$$

or more generally by using:

$$\mathbf{f}_{int} = \frac{\partial W_{int}}{\partial \mathbf{u}} \quad ; \quad \overline{\mathbf{K}} = \frac{\partial \mathbf{f}_{int}}{\partial \mathbf{u}}$$

Note that the stiffness matrix is a product of integration over volume (area for 2D solids).

Bilinear four node element

- Four nodes, 2 DOF's at node, 8 DOF's
- Linear interpolation for u_h, v_h
- Cannot describe pure bending - overly stiff in bending (spurious shear deformation)
- Suffers from shear locking

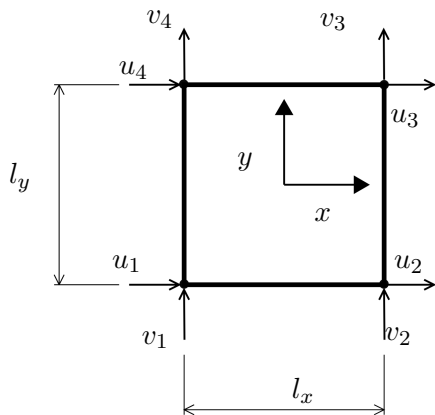
Basis functions for a four-node bilinear element (Q4):

$$\mathbf{p} = \{1, x, y, xy\} \quad (13)$$

Displacement field interpolations:

$$\begin{cases} u_h(x, y) := a_0 + a_1x + a_2y + a_3xy \\ v_h(x, y) := b_0 + b_1x + b_2y + b_3xy \end{cases} \quad (14)$$

Bilinear four node element



Example: shape functions for a four-node bilinear plane element

Basis functions for a four-node bilinear element (Q4):

$$\mathbf{p} = \{1, x, y, xy\} \quad (15)$$

Displacement field interpolations:

$$\begin{cases} u_h(x, y) := a_0 + a_1x + a_2y + a_3xy \\ v_h(x, y) := b_0 + b_1x + b_2y + b_3xy \end{cases}.$$

Displacement interpolation is same for the both displacements
 $a \Rightarrow$ same shape functions. Four unknowns, four eqs:

$$\begin{cases} u_h(-l_x/2, -l_y/2) := a_0 - \frac{1}{2}a_1l_x - \frac{1}{2}a_2l_y + \frac{1}{4}a_3l_xl_y = u_1 \\ u_h(l_x/2, -l_y/2) := a_0 + \frac{1}{2}a_1l_x - \frac{1}{2}a_2l_y - \frac{1}{4}a_3l_xl_y = u_2 \\ u_h(l_x/2, l_y/2) := a_0 + \frac{1}{2}a_1l_x + \frac{1}{2}a_2l_y + \frac{1}{4}a_3l_xl_y = u_3 \\ u_h(-l_x/2, l_y/2) := a_0 - \frac{1}{2}a_1l_x + \frac{1}{2}a_2l_y - \frac{1}{4}a_3l_xl_y = u_4 \end{cases}.$$

Example: shape functions for a four-node bilinear plane element

$$\begin{cases} u_h(-l_x/2, -l_y/2) := a_0 - \frac{1}{2}a_1l_x - \frac{1}{2}a_2l_y + \frac{1}{4}a_3l_xl_y = u_1 \\ u_h(l_x/2, -l_y/2) := a_0 + \frac{1}{2}a_1l_x - \frac{1}{2}a_2l_y - \frac{1}{4}a_3l_xl_y = u_2 \\ u_h(l_x/2, l_y/2) := a_0 + \frac{1}{2}a_1l_x + \frac{1}{2}a_2l_y + \frac{1}{4}a_3l_xl_y = u_3 \\ u_h(-l_x/2, l_y/2) := a_0 - \frac{1}{2}a_1l_x + \frac{1}{2}a_2l_y - \frac{1}{4}a_3l_xl_y = u_4 \end{cases}.$$

In a matrix form:

$$\underbrace{\begin{pmatrix} 1 & -1/2l_x & -1/2l_y & 1/4l_xl_y \\ 1 & 1/2l_x & -1/2l_y & -1/4l_xl_y \\ 1 & 1/2l_x & 1/2l_y & 1/4l_xl_y \\ 1 & -1/2l_x & 1/2l_y & -1/4l_xl_y \end{pmatrix}}_{\mathbf{A}} \underbrace{\begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix}}_{\mathbf{a}} = \underbrace{\begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix}}_{\mathbf{u}_u}. \quad (16)$$

Example: shape functions for a four-node bilinear plane element

By solving a_i 's:

$$\mathbf{a} = \mathbf{A}^{-1} \mathbf{u}_u \quad (17)$$

and substituting them into the Eq.(18):

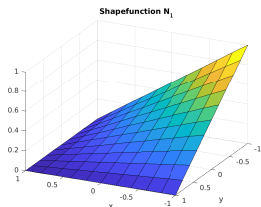
$$u_h = \mathbf{a}^T \mathbf{p} = \mathbf{A}^{-T} \mathbf{u}_u^T \mathbf{p}^T = \underbrace{\mathbf{A}^{-T} \mathbf{p}}_{\mathbf{N}} \mathbf{u}_u \quad (18)$$

where shape functions $\mathbf{N}$ can be written as

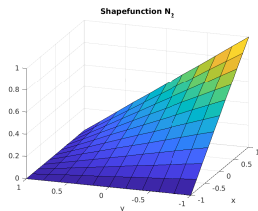
$$N_1 = \frac{(a-x)(b-y)}{4ab} ; N_2 = \frac{(a+x)(b-y)}{4ab} ; N_3 = \frac{(a+x)(b+y)}{4ab} ; N_4 = \frac{(a-x)(b+y)}{4ab}$$

where $a = l_x/2$ and $b = l_y/2$.

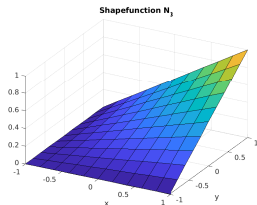
Example: shape functions for a four-node bilinear plane element



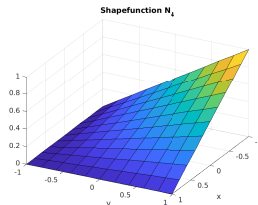
(a) $N_1(-1, -1) = 1$



(b) $N_2(1, -1) = 1$



(c) $N_3(1, 1) = 1$



(d) $N_4(-1, 1) = 1$

Strain energy

Displacement field $\mathbf{u}_h$ is interpolated as follows

Displacement field:

$$\underbrace{\begin{pmatrix} u_h \\ v_h \end{pmatrix}}_{\mathbf{u}_h} = \underbrace{\begin{pmatrix} N_1 & 0 & N_2 & 0 & N_3 & 0 & N_4 & 0 \\ 0 & N_1 & 0 & N_2 & 0 & N_3 & 0 & N_4 \end{pmatrix}}_{\mathbf{N}_m} \underbrace{\begin{pmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \\ u_4 \\ v_4 \end{pmatrix}}_{\mathbf{u}} \quad (19)$$

$$\mathbf{u}_h = \mathbf{N}_m \mathbf{u} \quad (20)$$

Strain energy and elemental stiffness matrix

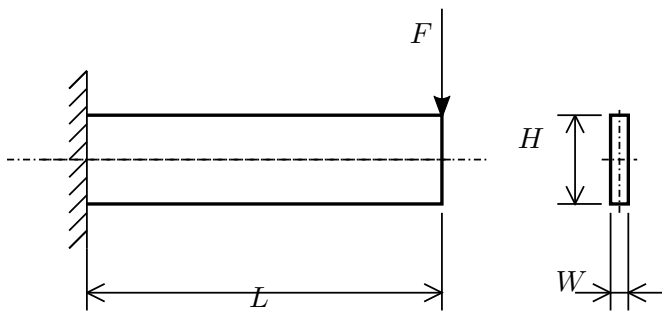
$$W_{int} = \frac{1}{2} \int_V \boldsymbol{\sigma} \boldsymbol{\varepsilon} \, dV = \frac{1}{2} \mathbf{u}^T l_z \underbrace{\int_x \int_y \mathbf{B}^T \mathbf{D} \mathbf{B} \, dy dx}_{\overline{\mathbf{K}}} \mathbf{u} \quad (21)$$

$$\overline{\mathbf{K}} = \int_V \mathbf{B}^T \mathbf{D} \mathbf{B} \, dV = l_z \int_x \int_y \mathbf{B}^T \mathbf{D} \mathbf{B} \, dy dx = l_z \int_{-l_x}^{l_x} \int_{-l_y}^{l_y} \mathbf{B}^T \mathbf{D} \mathbf{B} \, dy dx \quad (22)$$

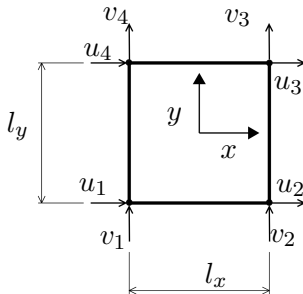
or more generally as:

$$\mathbf{f}_{int} = \frac{\partial W_{int}}{\partial \mathbf{u}} \quad ; \quad \overline{\mathbf{K}} = \frac{\partial \mathbf{f}_{int}}{\partial \mathbf{u}}$$

Example: Cantilever beam analyzed by Q4



Bilinear four node element

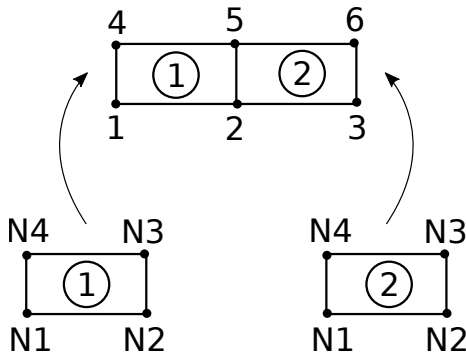


$$\underbrace{\begin{pmatrix} u_h \\ v_h \end{pmatrix}}_{\mathbf{u}_h} = \underbrace{\begin{pmatrix} N_1 & 0 & N_2 & 0 & N_3 & 0 & N_4 & 0 \\ 0 & N_1 & 0 & N_2 & 0 & N_3 & 0 & N_4 \end{pmatrix}}_{\mathbf{N}} \underbrace{\begin{pmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \\ u_4 \\ v_4 \end{pmatrix}}_{\mathbf{u}} \quad (23)$$

$$\mathbf{u}_h = \mathbf{N}\mathbf{u} \quad (24)$$

Example: Cantilever beam analyzed by Q4

Global nodal numbering



Stiffness matrices for a structure

Element 1 consists of nodes 1,2,5,4:

$$\overline{\mathbf{K}}^{(1)} = \begin{matrix} & \begin{matrix} u_1 & v_1 & u_2 & v_2 & u_5 & v_5 & u_4 & v_4 \end{matrix} \\ \left(\begin{matrix} k_{11}^{(1)} & k_{12}^{(1)} & k_{13}^{(1)} & k_{14}^{(1)} & k_{15}^{(1)} & k_{16}^{(1)} & k_{17}^{(1)} & k_{18}^{(1)} \\ k_{21}^{(1)} & k_{22}^{(1)} & k_{23}^{(1)} & k_{24}^{(1)} & k_{25}^{(1)} & k_{26}^{(1)} & k_{27}^{(1)} & k_{28}^{(1)} \\ k_{31}^{(1)} & k_{32}^{(1)} & k_{33}^{(1)} & k_{34}^{(1)} & k_{35}^{(1)} & k_{36}^{(1)} & k_{37}^{(1)} & k_{38}^{(1)} \\ k_{41}^{(1)} & k_{42}^{(1)} & k_{43}^{(1)} & k_{44}^{(1)} & k_{45}^{(1)} & k_{46}^{(1)} & k_{47}^{(1)} & k_{48}^{(1)} \\ k_{51}^{(1)} & k_{52}^{(1)} & k_{53}^{(1)} & k_{54}^{(1)} & k_{55}^{(1)} & k_{56}^{(1)} & k_{57}^{(1)} & k_{58}^{(1)} \\ k_{61}^{(1)} & k_{62}^{(1)} & k_{63}^{(1)} & k_{64}^{(1)} & k_{65}^{(1)} & k_{66}^{(1)} & k_{67}^{(1)} & k_{68}^{(1)} \\ k_{71}^{(1)} & k_{72}^{(1)} & k_{73}^{(1)} & k_{74}^{(1)} & k_{75}^{(1)} & k_{76}^{(1)} & k_{77}^{(1)} & k_{78}^{(1)} \\ k_{81}^{(1)} & k_{82}^{(1)} & k_{83}^{(1)} & k_{84}^{(1)} & k_{85}^{(1)} & k_{86}^{(1)} & k_{87}^{(1)} & k_{88}^{(1)} \end{matrix} \right) & \begin{matrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_5 \\ v_5 \\ u_4 \\ v_4 \end{matrix} \end{matrix}$$

Stiffness matrices for a structure

Element 2 consists of nodes 2,3,6,5:

$$\overline{K}^{(2)} = \begin{matrix} & \begin{matrix} u_2 & v_2 & u_3 & v_3 & u_6 & v_6 & u_5 & v_5 \end{matrix} \\ \left(\begin{array}{cccccccc} k_{11}^{(2)} & k_{12}^{(2)} & k_{13}^{(2)} & k_{14}^{(2)} & k_{15}^{(2)} & k_{16}^{(2)} & k_{17}^{(2)} & k_{18}^{(2)} \\ k_{21}^{(2)} & k_{22}^{(2)} & k_{23}^{(2)} & k_{24}^{(2)} & k_{25}^{(2)} & k_{26}^{(2)} & k_{27}^{(2)} & k_{28}^{(2)} \\ k_{31}^{(2)} & k_{32}^{(2)} & k_{33}^{(2)} & k_{34}^{(2)} & k_{35}^{(2)} & k_{36}^{(2)} & k_{37}^{(2)} & k_{38}^{(2)} \\ k_{41}^{(2)} & k_{42}^{(2)} & k_{43}^{(2)} & k_{44}^{(2)} & k_{45}^{(2)} & k_{46}^{(2)} & k_{47}^{(2)} & k_{48}^{(2)} \\ k_{51}^{(2)} & k_{52}^{(2)} & k_{53}^{(2)} & k_{54}^{(2)} & k_{55}^{(2)} & k_{56}^{(2)} & k_{57}^{(2)} & k_{58}^{(2)} \\ k_{61}^{(2)} & k_{62}^{(2)} & k_{63}^{(2)} & k_{64}^{(2)} & k_{65}^{(2)} & k_{66}^{(2)} & k_{67}^{(2)} & k_{68}^{(2)} \\ k_{71}^{(2)} & k_{72}^{(2)} & k_{73}^{(2)} & k_{74}^{(2)} & k_{75}^{(2)} & k_{76}^{(2)} & k_{77}^{(2)} & k_{78}^{(2)} \\ k_{81}^{(2)} & k_{82}^{(2)} & k_{83}^{(2)} & k_{84}^{(2)} & k_{85}^{(2)} & k_{86}^{(2)} & k_{87}^{(2)} & k_{88}^{(2)} \end{array} \right) & \begin{matrix} u_2 \\ v_2 \\ u_3 \\ v_3 \\ u_6 \\ v_6 \\ u_5 \\ v_5 \end{matrix} \end{matrix}$$

Global stiffness matrix is a size of 12×12 . Nodes 2 and 5 are common nodes. Nodal displacements are fixed at nodes 1 and 4.

$$K_{Gred} = \begin{pmatrix} k_{33}^{(1)} + k_{11}^{(2)} & k_{34}^{(1)} + k_{12}^{(2)} & k_{13}^{(2)} & k_{14}^{(2)} & k_{35}^{(1)} + k_{17}^{(2)} & k_{36}^{(1)} + k_{18}^{(2)} & k_{1,5}^{(2)} & k_{16}^{(2)} & u_2 \\ k_{43}^{(1)} + k_{21}^{(2)} & k_{44}^{(1)} + k_{22}^{(2)} & k_{23}^{(2)} & k_{24}^{(2)} & k_{45}^{(1)} + k_{27}^{(2)} & k_{46}^{(1)} + k_{28}^{(2)} & k_{25}^{(2)} & k_{26}^{(2)} & u_2 \\ k_{31}^{(2)} & k_{32}^{(2)} & k_{33}^{(2)} & k_{34}^{(2)} & k_{37}^{(2)} & k_{38}^{(2)} & k_{35}^{(2)} & k_{36}^{(2)} & u_3 \\ k_{41}^{(2)} & k_{42}^{(2)} & k_{43}^{(2)} & k_{44}^{(2)} & k_{47}^{(2)} & k_{48}^{(2)} & k_{45}^{(2)} & k_{46}^{(2)} & u_3 \\ k_{53}^{(1)} + k_{71}^{(2)} & k_{54}^{(1)} + k_{72}^{(2)} & k_{73}^{(2)} & k_{74}^{(2)} & k_{55}^{(1)} + k_{77}^{(2)} & k_{56}^{(1)} + k_{78}^{(2)} & k_{75}^{(2)} & k_{76}^{(2)} & u_5 \\ k_{63}^{(1)} + k_{81}^{(2)} & k_{64}^{(1)} + k_{82}^{(2)} & k_{83}^{(2)} & k_{84}^{(2)} & k_{65}^{(1)} + k_{87}^{(2)} & k_{66}^{(1)} + k_{88}^{(2)} & k_{85}^{(2)} & k_{86}^{(2)} & u_5 \\ k_{51}^{(2)} & k_{52}^{(2)} & k_{53}^{(2)} & k_{54}^{(2)} & k_{57}^{(2)} & k_{58}^{(2)} & k_{55}^{(2)} & k_{56}^{(2)} & u_6 \\ k_{61}^{(2)} & k_{62}^{(2)} & k_{63}^{(2)} & k_{64}^{(2)} & k_{67}^{(2)} & k_{68}^{(2)} & k_{65}^{(2)} & k_{66}^{(2)} & u_6 \end{pmatrix}$$

Force vector and solution

After taking account for boundary conditions

$u_1 = v_1 = u_4 = v_4 = 0$, global force vector can be written as:

$$\mathbf{f}_{G_{red}} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ -F/2 \\ 0 \\ 0 \\ 0 \\ -F/2 \end{pmatrix} \begin{matrix} u_2 \\ v_2 \\ u_3 \\ v_3 \\ u_5 \\ v_5 \\ u_6 \\ v_6 \end{matrix}$$

$$\frac{d^2v}{dx^2} = -\frac{M(x)}{EI(x)} - \frac{\kappa_{xy}q(x)}{GA} \quad (25)$$

Maximum deflection for the cantilever beam under applied force at the free end can be found as:

$$\delta_{max} = \underbrace{\frac{FL^3}{3EI}}_{\text{bending}} + \kappa_{xy} \underbrace{\frac{FL}{GA}}_{\text{shear}} \quad (26)$$

where κ_{xy} is a shear correction factor (needed to use in the beams based on the first-order shear deformation theory (FSDT)). It can be approximated to be 6/5 for a rectangular cross-section.